

Assignment 3 — Extending the “Firefox New Tab / News Cards” Application

Enhance your existing “Firefox New Tab / News Cards” project by integrating **React Router**, **Axios**, **Context**, and **Redux Toolkit**. This week focuses on routing, data fetching, and state management.

Tech Stack

- React (Vite + TypeScript)
- Tailwind CSS
- shadcn/ui
- React Router DOM
- Axios
- Redux Toolkit
- React Context API

Assignment Objectives

1. Routing (React Router)

- Add multiple pages:
 - `/` — Home page (existing news cards grid)
 - `/news/:id` — Detailed news page showing the selected story’s full content.
- Add navigation support to switch between pages.
- Clicking on any news card should navigate to its detail page.
- Use **React Router DOM v6** with clean and modular routing setup.

2. Axios Integration

- Replace static JSON with real or mock API data.
- Use **Axios** to fetch news data from:
 - [NewsAPI.org](https://newsapi.org/) (requires free API key)
 - or **Mock API** via Mocky.io / [JSONPlaceholder](https://jsonplaceholder.typicode.com/) / [MockAPI.io](https://mockapi.io/)
- Create a service layer (`api/newsService.ts`) to handle API calls.
- Handle loading and error states gracefully with UI feedback.

3. Global State Management

Implement two state management layers:

- **Context API:** For app-wide settings (like theme or background image).
- **Redux Toolkit:** For managing fetched news data, including:
 - Storing the news list.
 - Selecting a specific news item for detail view.
- Demonstrate how Redux Toolkit and Context can coexist and serve different purposes.

4. UI & Interactivity Enhancements

- Improve the UX with a clean transition from grid → detail view.
- Add a “Back to Home” button on the detail page.
- Add a theme/background change feature using Context.
- Use shadcn/ui components for consistency (e.g., Button, Card, Spinner).

Milestones

Milestone 1 – Routing Setup

- Install and configure `react-router-dom`.
- Create routes for Home and News Detail.
- Implement navigation.

Milestone 2 – Axios Integration

- Replace static data with Axios calls.
- Display fetched news dynamically.
- Implement loading and error UI states.

Milestone 3 – Redux Toolkit

- Set up Redux store and slice for news data.
- Fetch and store data using `createAsyncThunk`.
- Access Redux state in components.

Milestone 4 – Context API

- Implement Context for background/theme management.
- Create toggle functionality.
- Apply theme across the app.

Learning Outcomes

By completing this assignment, you'll:

- Understand and implement client-side routing using React Router.
- Learn API integration and async data fetching with Axios.
- Use Redux Toolkit for scalable state management.
- Apply Context for lightweight global state.
- Practice modular React architecture for maintainable apps.

GitHub Guidelines

- Repository: Continue in `firefox-newtab-react`
- Branch strategy:
 - `feature/routing`
 - `feature/axios`
 - `feature/redux`
 - `feature/context`

Submission Requirements

- Functional app build (`yarn dev`)
- Updated README with:
 - Routes overview
 - API details
 - State management explanation (Redux + Context)
- Screenshots of all major features
- Proper commit history following Git best practices